



NSS College of Engineering,
Department of Electrical and Electronics Engineering

ANN BASED ACTIVE POWER FILTER FOR ADAPTIVE HARMONIC COMPENSATION

PROJECT COORDINATOR

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Introduction

- Nonlinear loads → Harmonic Current
- SAPF can compensate harmonic current
- AI based APF can adapt to nonlinear conditions

Literature Review

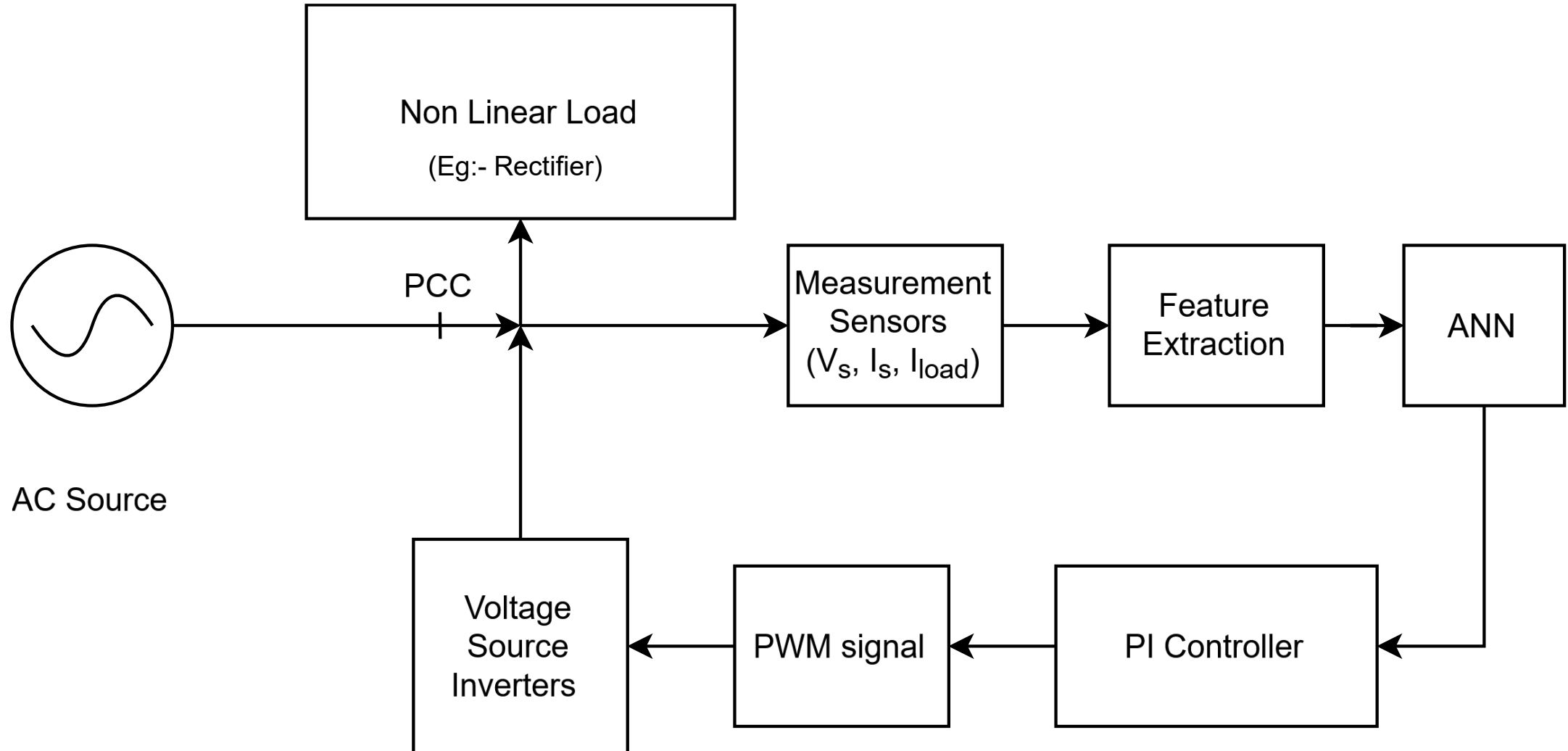
Sl. No.	Author	Title	Key contribution	Research gap / limitation
1	Flores-Garrido et al. (2021)	Nonlinear Loads Compensation Using a Shunt Active Power Filter Controlled by Feedforward Neural Networks	MLP-based APF reference generation; MATLAB + experimental validation; 100k training patterns	Large training set; focuses on effective ANN operation rather than minimum data requirement
2	Janpong et al. (2021)	Harmonic Detection for Shunt Active Power Filter Using ADALINE Neural Network	ADALINE reference-current generation; DSP/HIL implementation	Neural approach is validated, but model/data-efficiency trade-off is not the main focus
3	Deshpande et al. (2022)	Analysis of Levenberg-Marquardt-ANN Based Reference Current Generation for Control of SAPF	ANN reference-current generation under nonlinear/dynamic/unbalanced loads	Focuses on ANN performance; systematic data/model reduction is not the objective
4	Muneer et al. (2023)	Optimal Machine-Learning-Based Controller for SAPF by Auto Machine Learning	AutoML across >15 ML models and 110 hyperparameters; experimental prototype	Optimizes model selection; our focus is specifically data/model economy + unseen-load evaluation
5	Govind et al. (2024)	Simulation and Real Time Implementation of SAPF Using Adaptive Neural Network Topology	ANN/ADALINE APF; MATLAB/Simulink + DSP prototype; dynamic loads	Demonstrates real-time ANN APF; resource/data efficiency is not the central research question

Problem Statement

Limitations of conventional APF

- Fixed control strategy
- Limited adaptability to changing loads
- Parameter tuning is required (K_p and K_i of PI controller)

Proposed System



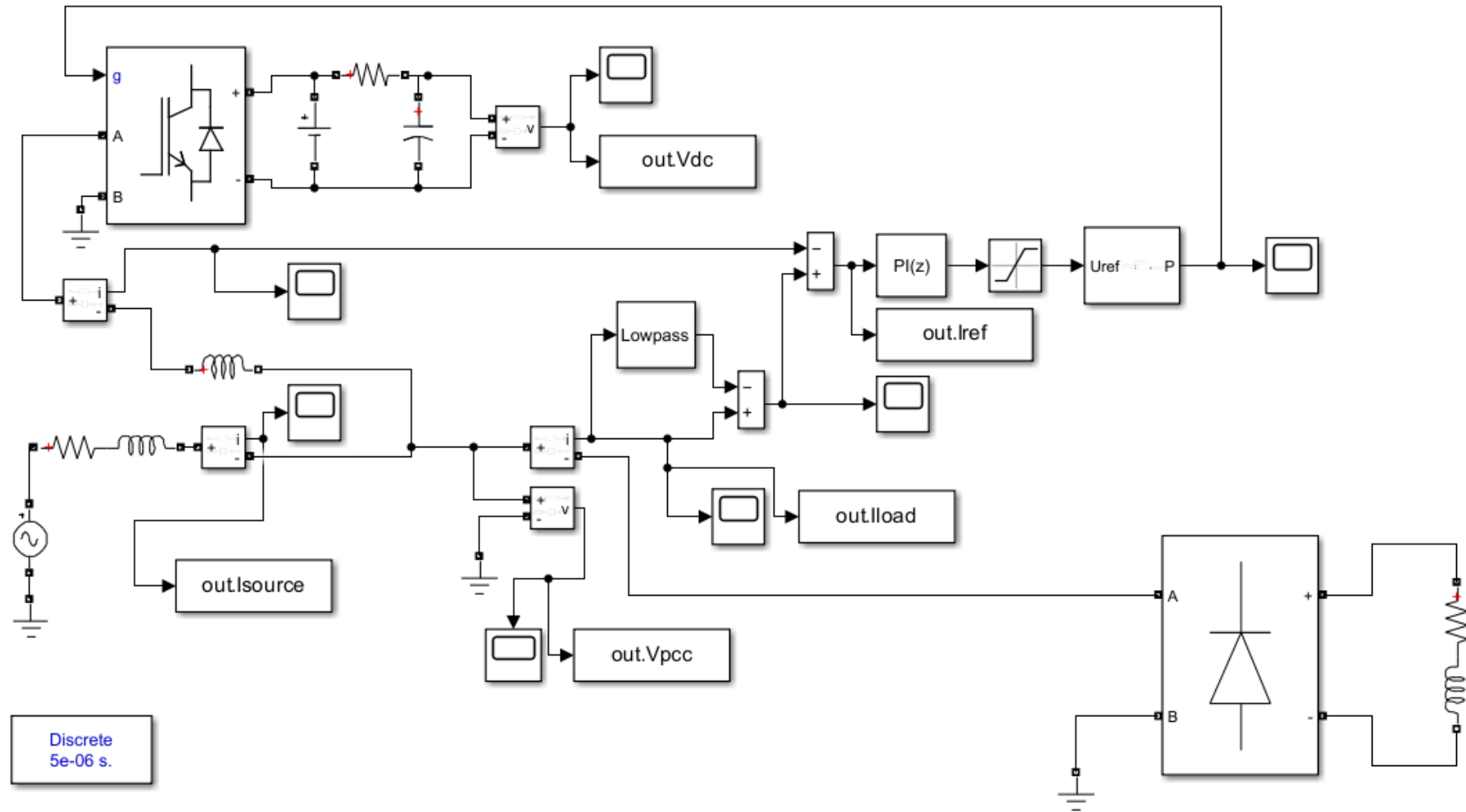
Proposed Methodology



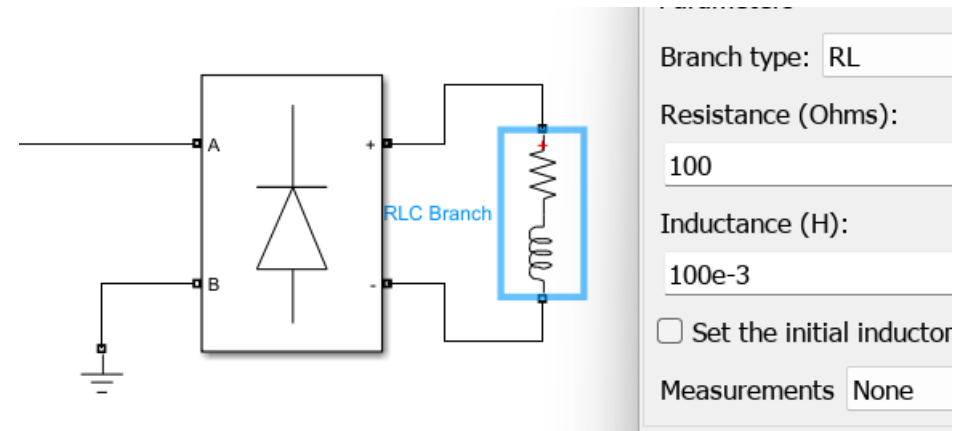
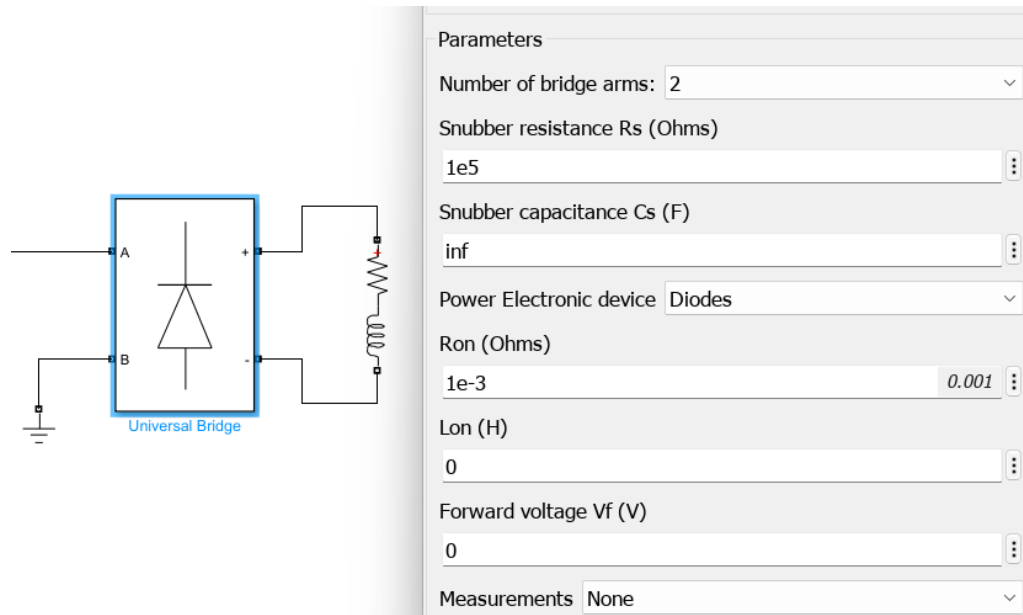
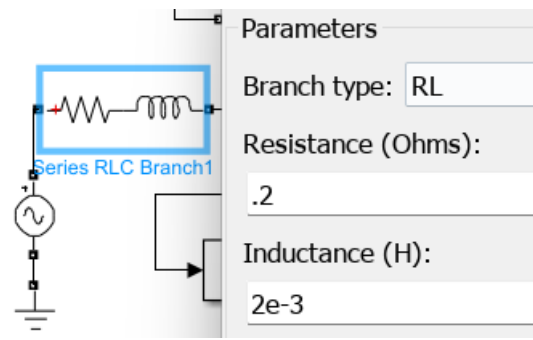
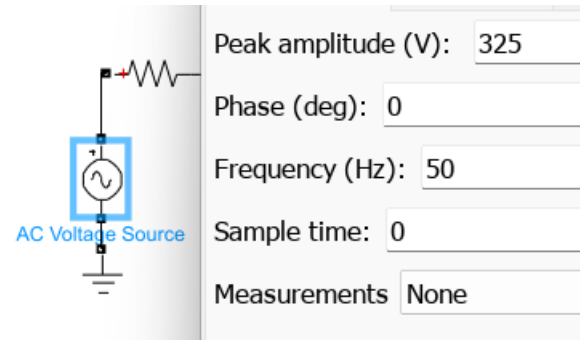
Feedback from zeroth review

- Specify the load used
- Refine the contents in slides
- What's the use of AI?

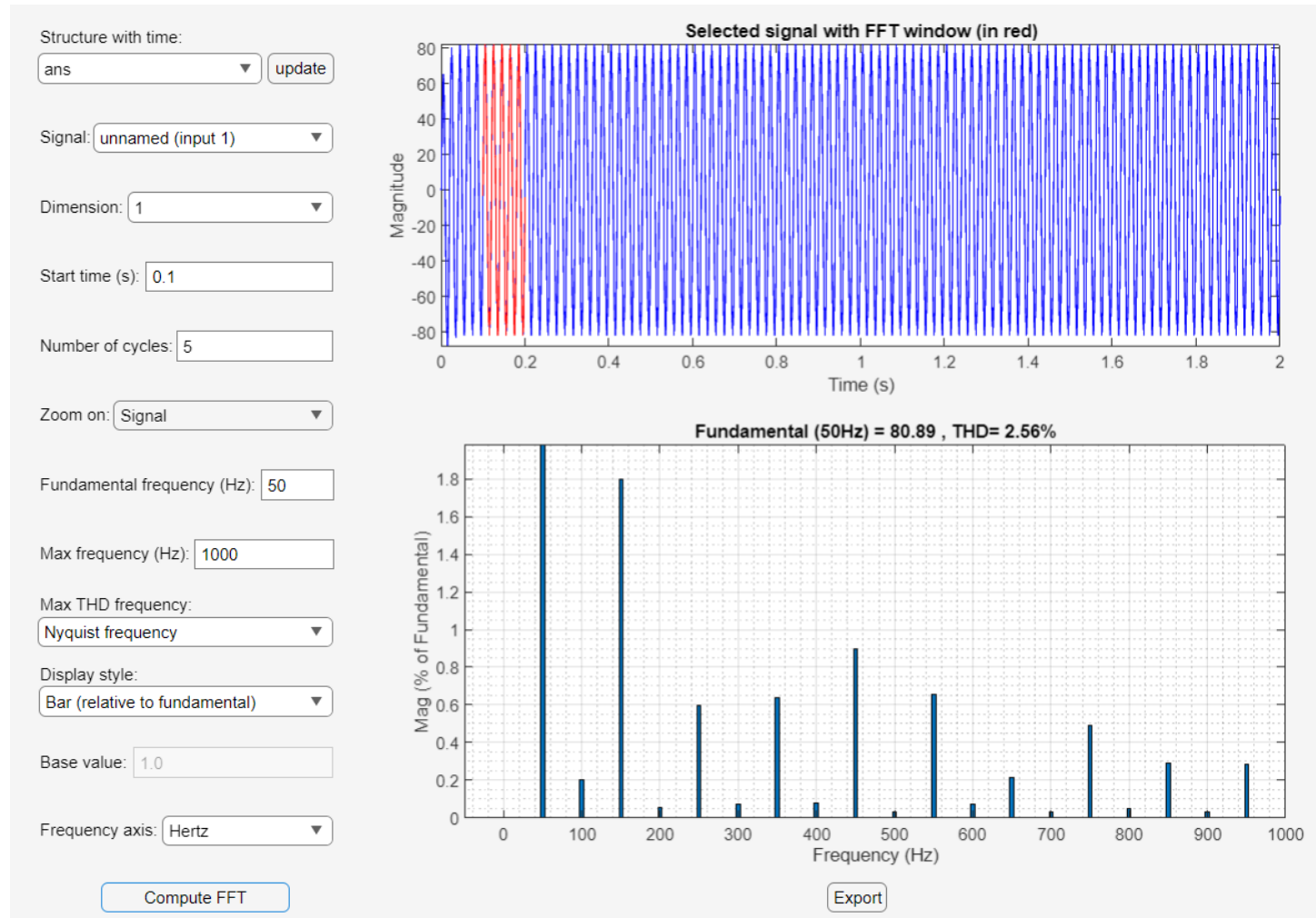
Conventional APF



Simulation Conditions



THD of Conventional APF

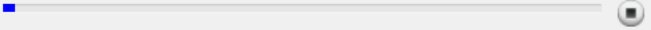


Dataset Generation

```
whos
out
Vpcc    = out.Vpcc;
ILoad   = out.ILoad;
Vdc     = out.Vdc;
Iref     = out.Iref;
Isorce  = out.Isorce;
whos
save TrainingData.mat Vpcc ILoad Vdc Iref Isorce
load TrainingData.mat
Vpcc_data = Vpcc.signals.values;
ILoad_data = ILoad.signals.values;
Vdc_data = Vdc.signals.values;
Iref_data = Iref.signals.values;
Isorce_data = Isorce.signals.values;
size(Vpcc_data)
size(ILoad_data)
size(Vdc_data)
size(Iref_data)
size(Isorce_data)
Iref_data = squeeze(Iref_data);
inputs = [Vpcc_data(:)'; ILoad_data(:)'; Vdc_data(:)'];
targets = Iref_data(:)';
save ANN_Dataset.mat inputs targets
train_ann
```

Neural Network Training (13-Sep-2026 15:35:25)

Network Diagram

Training Results
Epoch: 18 of 1000


Training Progress

Unit	Initial Value	Current Value	Target Value	
Epoch	0	18	1000	
Elapsed Time	-	00:00:13	-	
Performance	1.05e+05	61.6	0	
Gradient	1.88e+05	49.3	1e-07	
Mu	0.001	1	1e+10	
Validation Checks	0	0	6	

Training Algorithms
Data Division: Random dividerand
Training: Levenberg-Marquardt trainlm
Performance: Mean Squared Error mse
Calculations: MEX

Training Plots

Performance

Training State

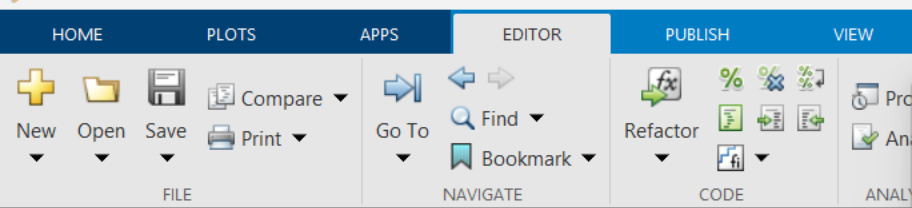
Error Histogram

Regression

Fit

ANN Training Script

```
train_ann.m
1  clear;
2  clc;
3
4  load ANN_Dataset.mat
5
6  net = fitnet(20);      % 20 hidden neurons
7
8  net.divideParam.trainRatio = 0.7;
9  net.divideParam.valRatio   = 0.15;
10 net.divideParam.testRatio  = 0.15;
11
12 [net,tr] = train(net, inputs, targets);
13
14 outputs = net(inputs);
15
16 figure;
17 plotperform(tr);
18
19 figure;
20 plotregression(targets, outputs);
21
22 figure;
23 plot(targets(1:2000), 'b');
24 hold on;
25 plot(outputs(1:2000), 'r');
26 legend('Target', 'ANN Output');
27 grid on;
28
29 save trainedANN.mat net
```



Current Folder: D:\Projects & Works\APF - Final Year Project

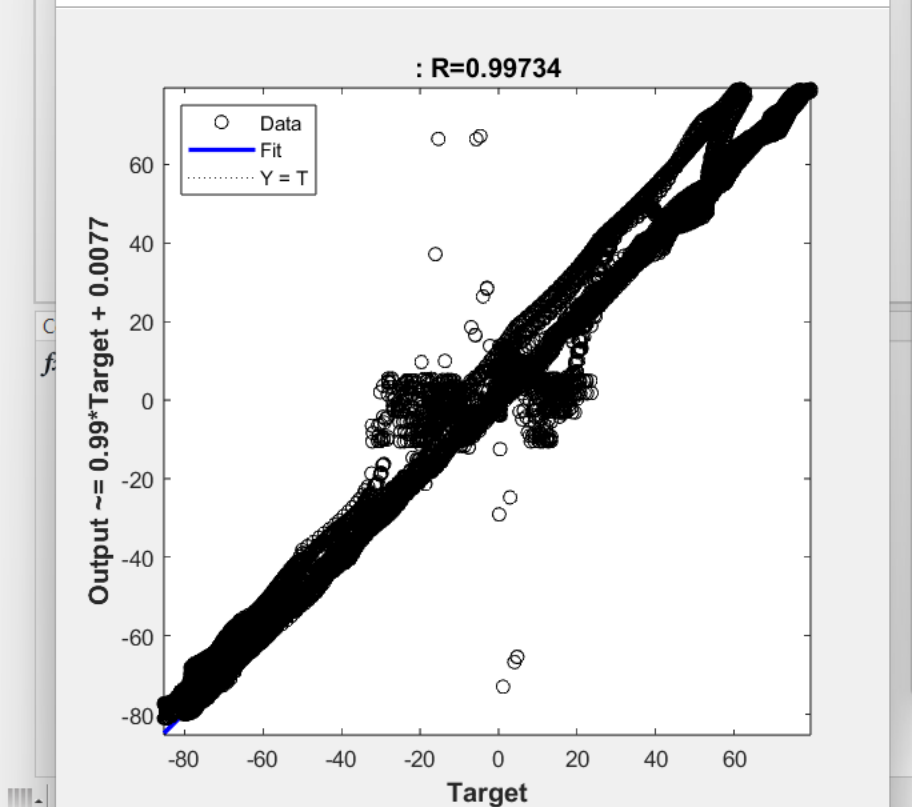
Editor - train_ann.m

exp1.m exp1_2.m stability1.m stability2.m train_ann.m

```
1 clear;
2 clc;
3
4 load ANN_Dataset.mat
5
6 net = fitnet(20); % 20 hidden neurons
```

Regression (plotregression)

File Edit View Insert Tools Desktop Window Help



Neural Network Training (04-Aug-2026 21:10:46)

Network Diagram

Training Results

Training finished: Met validation criterion ✓

Training Progress

Unit	Initial Value	Stopped Value	Target Value
Epoch	0	516	1000
Elapsed Time	-	00:05:05	-
Performance	7.49e+04	16.1	0
Gradient	1.3e+05	0.268	1e-07
Mu	0.001	0.001	1e+10
Validation Checks	0	6	6

Training Algorithms

Data Division: Random dividerand

Training: Levenberg-Marquardt trainlm

Performance: Mean Squared Error mse

Calculations: MEX

Training Plots

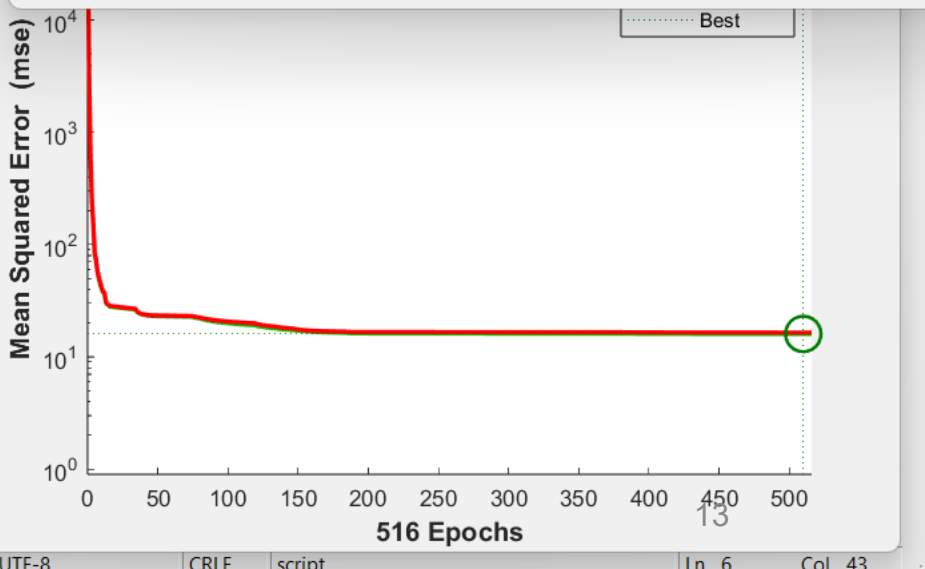
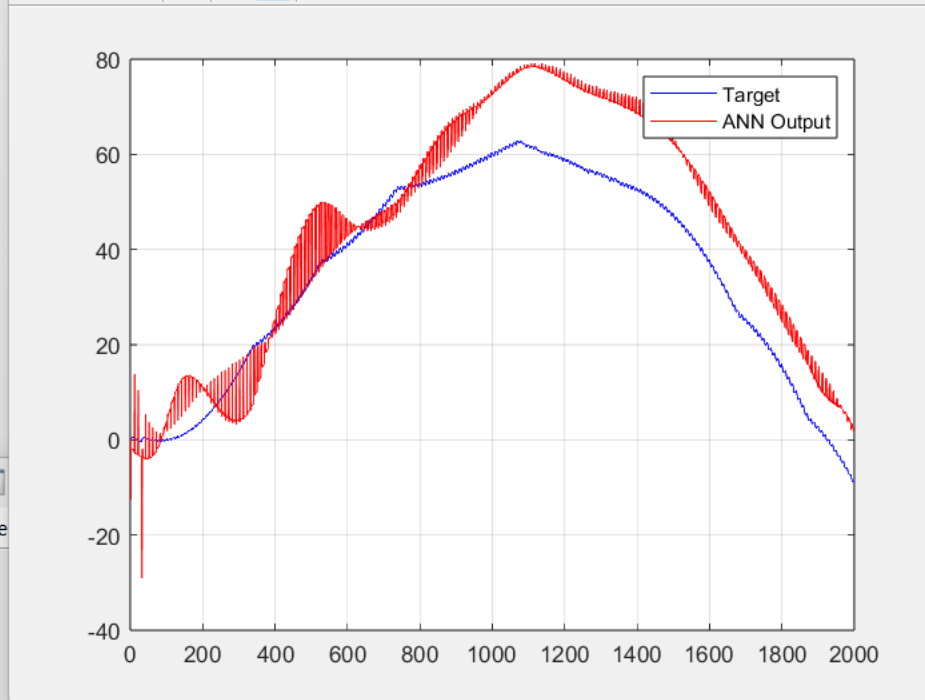
Performance Training State

Error Histogram Regression

Fit

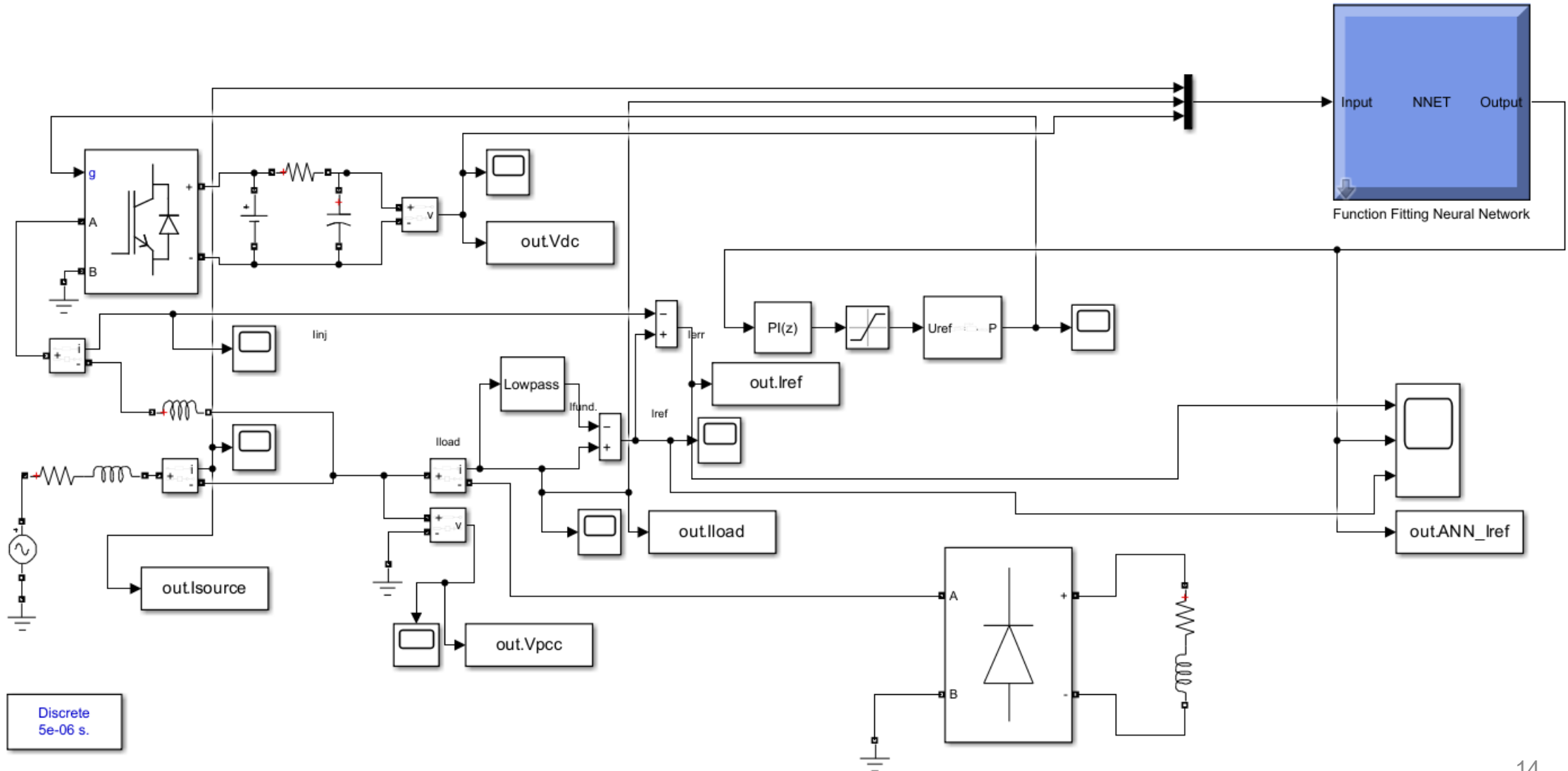
Figure 3

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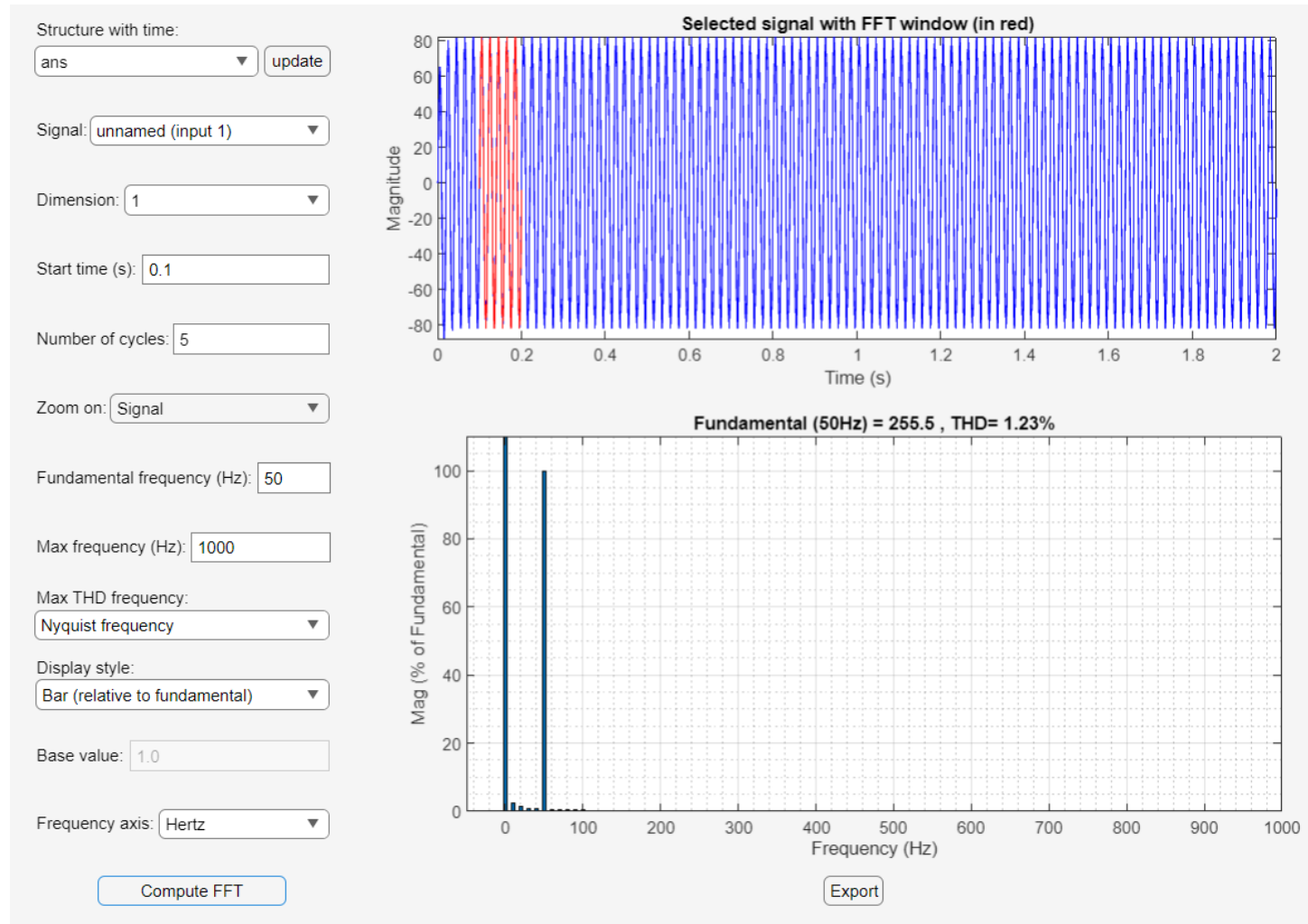


ANN Based APF

```
train_ann
save('ANN_Controller.mat','net');
gensim(net)
```



THD of ANN Based APF



Objectives

- To Reduce THD
- Learns the relationship between operating conditions and required compensation
- Improved harmonic compensation under varying conditions

Thank You